MAS508-G Control Theory
Lecture 3: State Space Design

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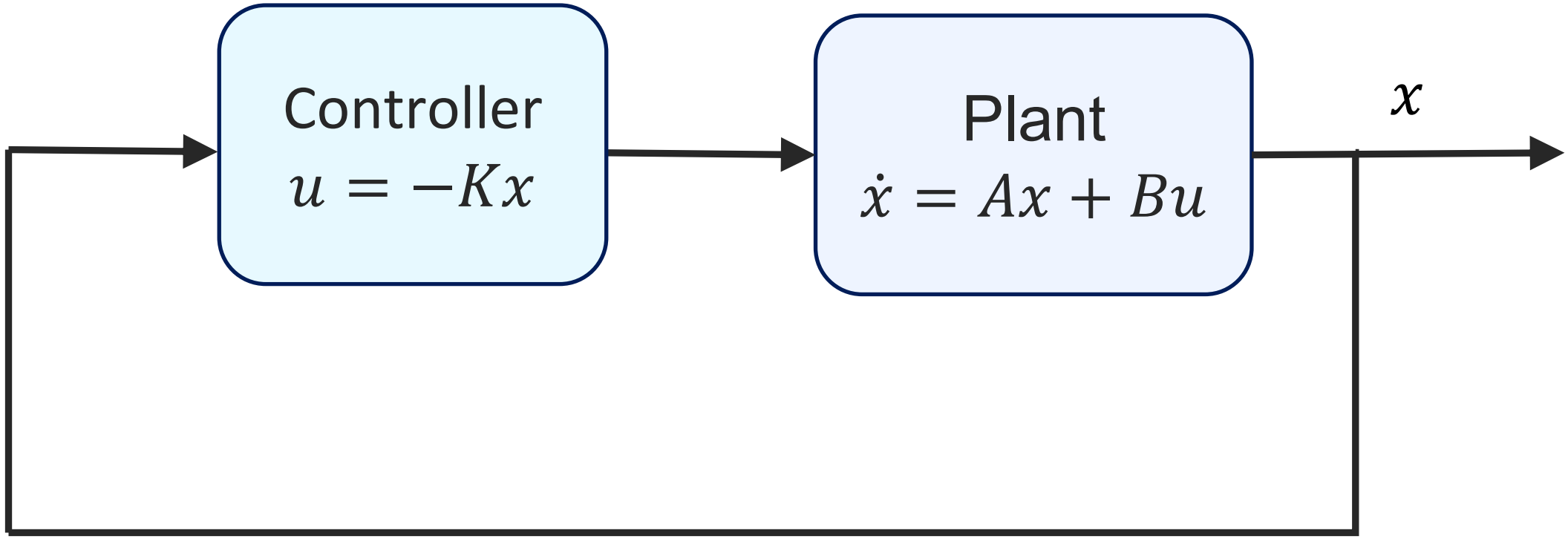
L3. State Space Design

a) Time Response
b) Controllability
c) Observability
d) Pole Placement Control
e) Observer Design
f) Output Feedback Control
g) Optimal Control - Linear Quadratic Control (LQR)
h) Optimal Control – LQG

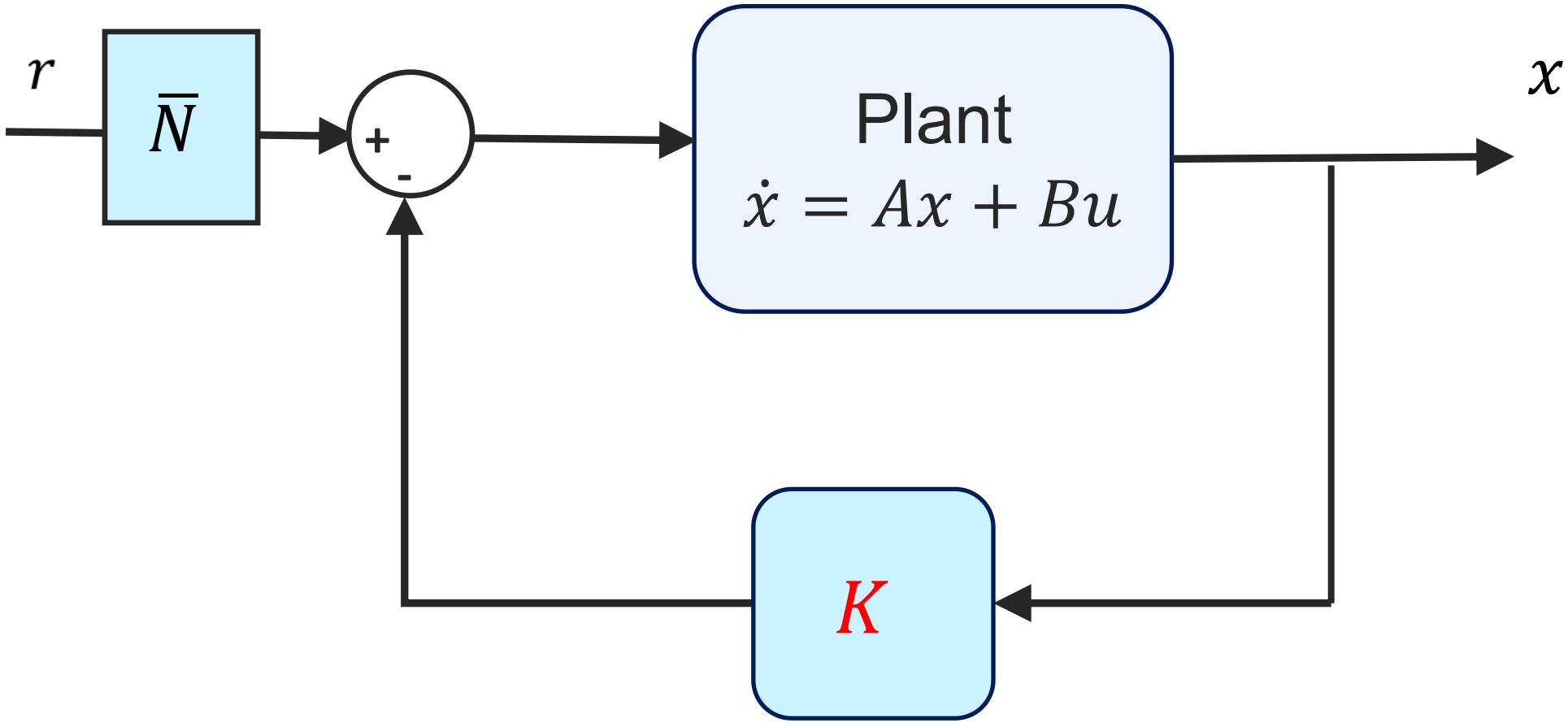
Textbook, “Control Systems Engineering”, Chapters 4 &12

L3g. Optimal Control
- Linear Quadratic Control (LQR)

State Feedback Control $u = -Kx$



State Feedback Control $u = -Kx + \bar{N}r$



Pole Placement Design to Find $\boldsymbol{K}$

- For a system with the state-space model:

$$\dot{x} = Ax + Bu, \qquad y = Cx$$

- Pole-placement finds a state-feedback control law:

$$u = -Kx$$

- That brings the poles of the closed-loop system to desired locations

$$\dot{x} = (A - BK)x$$

- Such as the characteristic equation $\det(sI - (A - BK)) = 0$ matches the desired characteristic equation. (Desired poles)

Pole Placement

- For a system with the state-space model:

$$\dot{x} = Ax + Bu, \qquad y = Cx$$

- Pole-placement find the gain $\boldsymbol{K}$ in state-feedback control :

$$u = -Kx$$

- That brings the poles of the closed-loop system to desired locations

$$\dot{x} = (A - BK)x$$

- Such that the characteristic equation of $(A - BK)$ matches the desired characteristic equation. (Desired poles $p_1, p_2, \dots p_n$)

$$\det(s\boldsymbol{I} - (A - BK)) = (s - p_1)(s - p_2)\cdots(s - p_n).$$

Example

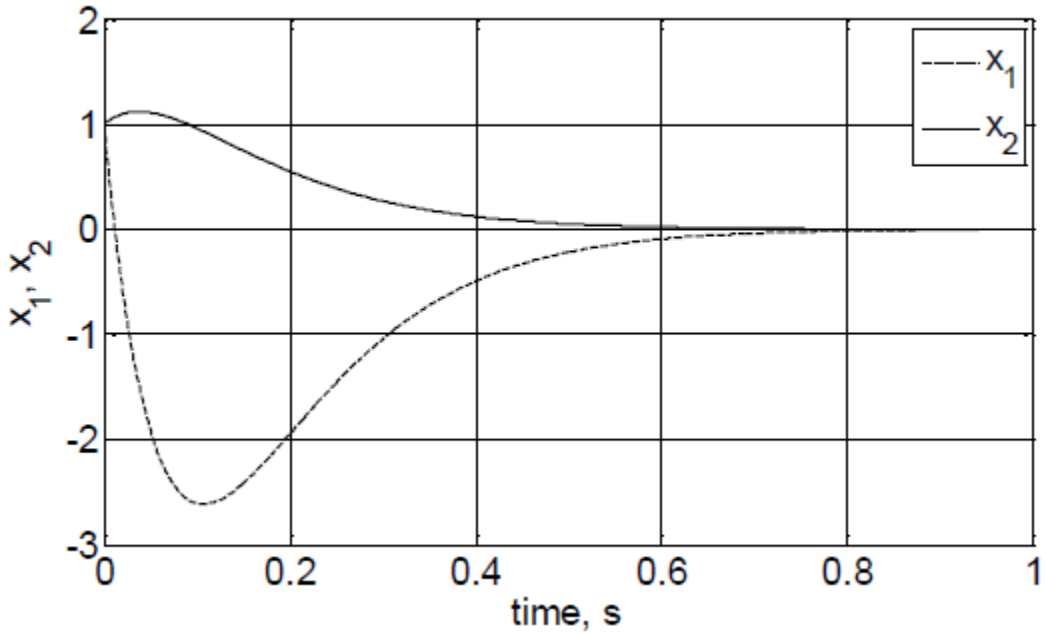
$$\dot{x}(t)=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}x(t)+\begin{bmatrix}1\\ 0\end{bmatrix}u(t),\ y(t)=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}x(t)$$

Design a feedback control $u=-Kx$ to

Place the poles at -10 and -11.

$K=[26\ \ 72];$

```
A=[1 2;3 4]; B=[1;0];
rank(ctrb(A,B))
K=place(A,B,[-10 -11]);
eig(A-B*K),
```



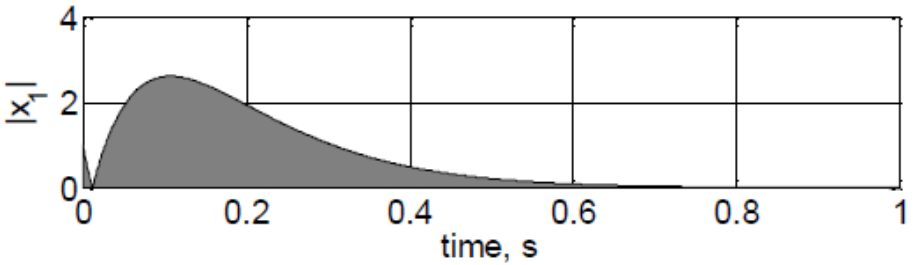
Example

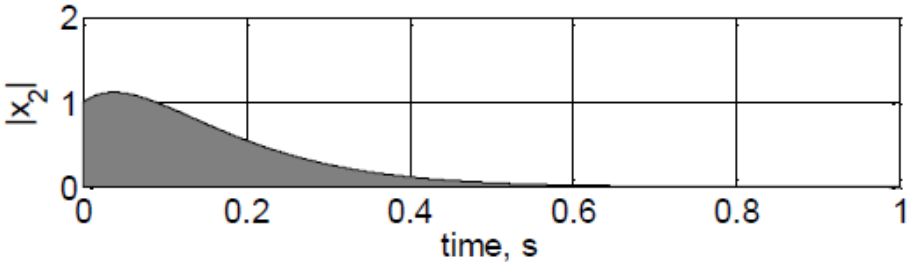
In order to quantify the phrase “converge to zero as fast as possible” we can use various performance indexes like:

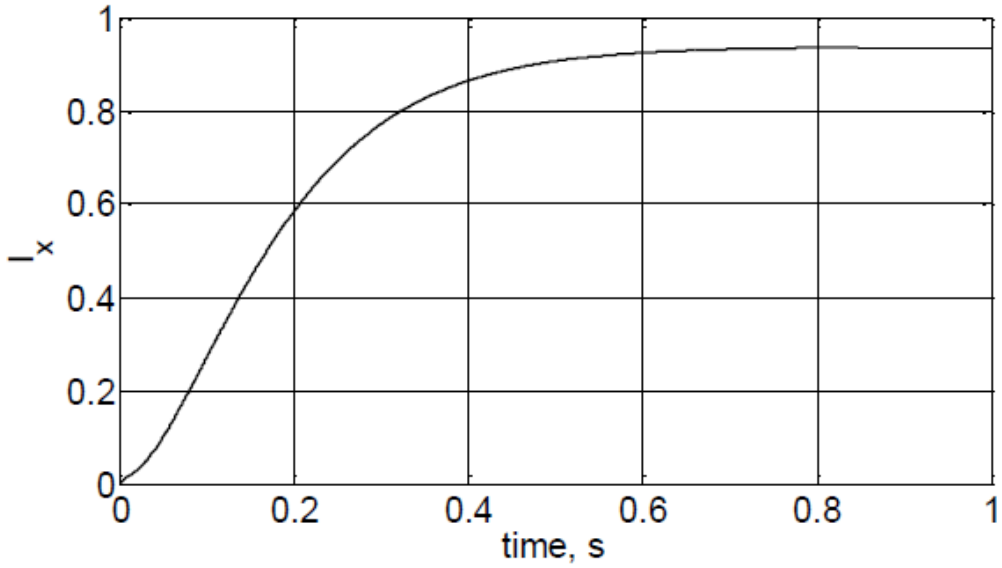
$$I_x = \int (|x_1| + |x_2|)\,\mathrm{dt}$$

This index effectively tries to minimize the area under the curves produced by $x1$, $x2$. Obviously the faster the system, the smaller the above areas and hence the smaller the performance index, in this case the index has as a final value

$$I_x = 0.94.$$







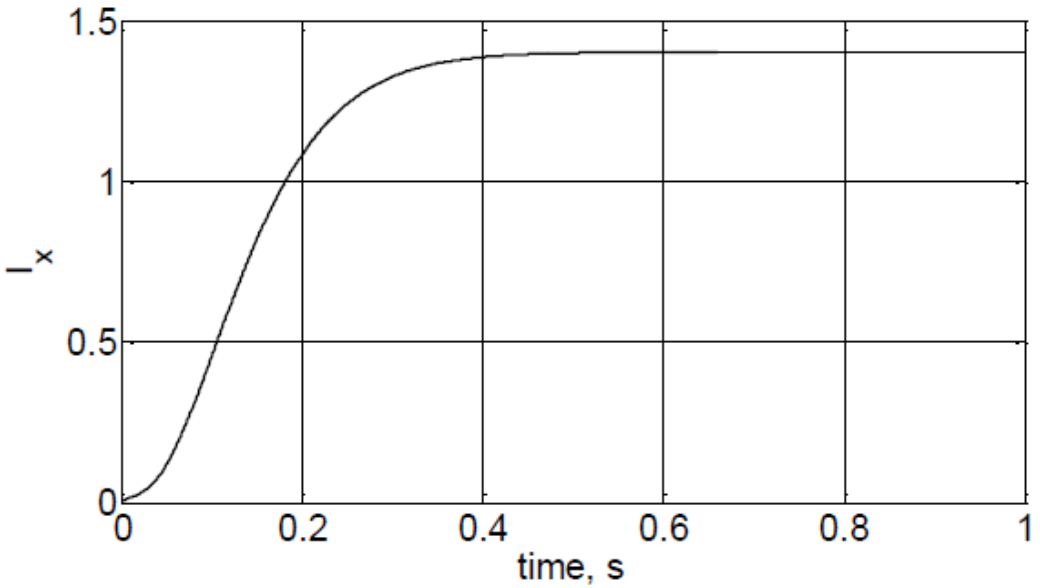
Example

In order to quantify the phrase “converge to zero as fast as possible” we can use another performance indexes:

$$I_x = \int (x_1^2 + x_2^2)\,\mathrm{dt}$$

In this case the index has as a final value

$I_x = 1.40.$



Example - Case 1

$$\dot{x}(t)=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}x(t)+\begin{bmatrix}1\\ 0\end{bmatrix}u(t),\ y(t)=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}x(t)$$

$K = [26\ \ 72]$

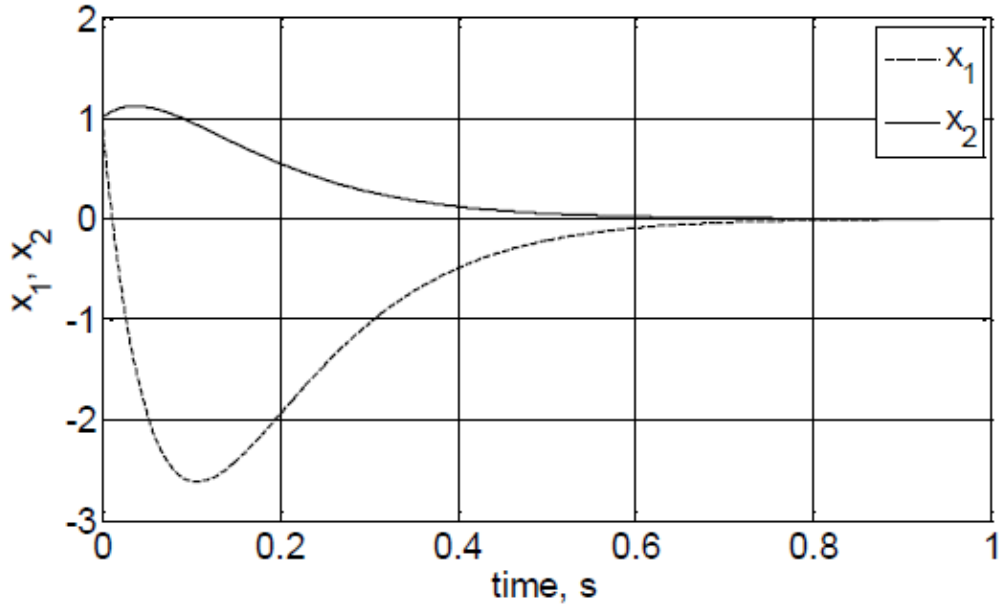
The feedback control $u = -Kx$ to place the poles at -10 and -11.

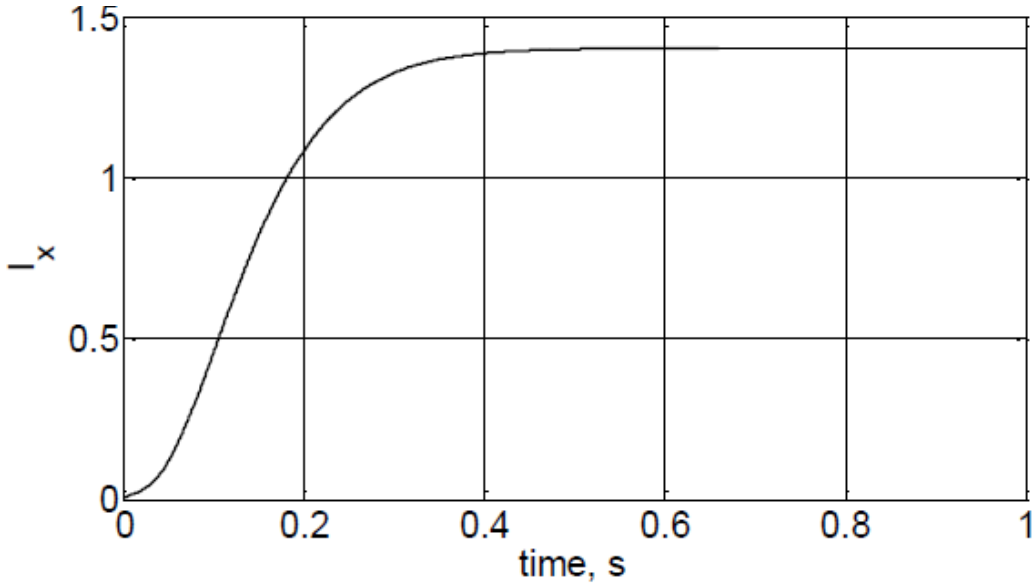
In order to quantify the phrase "converge to zero as fast as possible" we can use performance indexes:

$$I_x = \int (x_1^2 + x_2^2)\,\mathrm{dt}$$

In this case the index has as a final value

$\boldsymbol{I_x = 1.40}$.





Example - Case 2

$$\dot{x}(t)=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}x(t)+\begin{bmatrix}1\\ 0\end{bmatrix}u(t),\ y(t)=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}x(t)$$

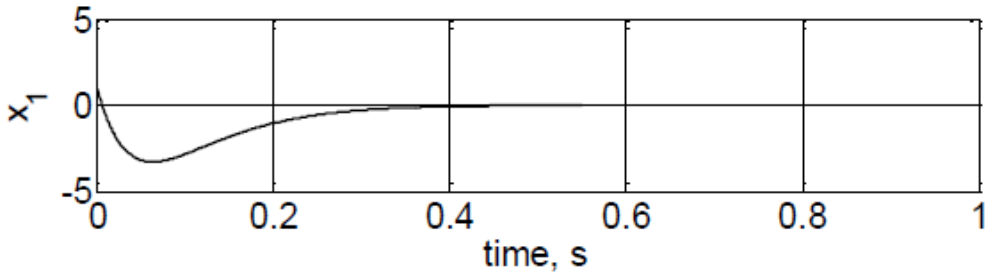
$K = [40 \quad 154]$

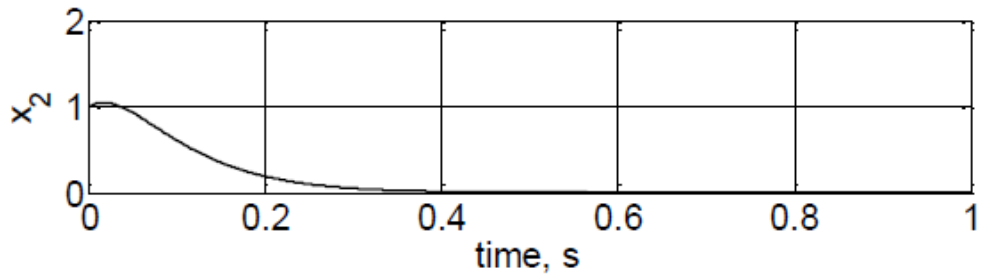
The feedback control $u = -Kx$ to place the poles at -15 and -20.

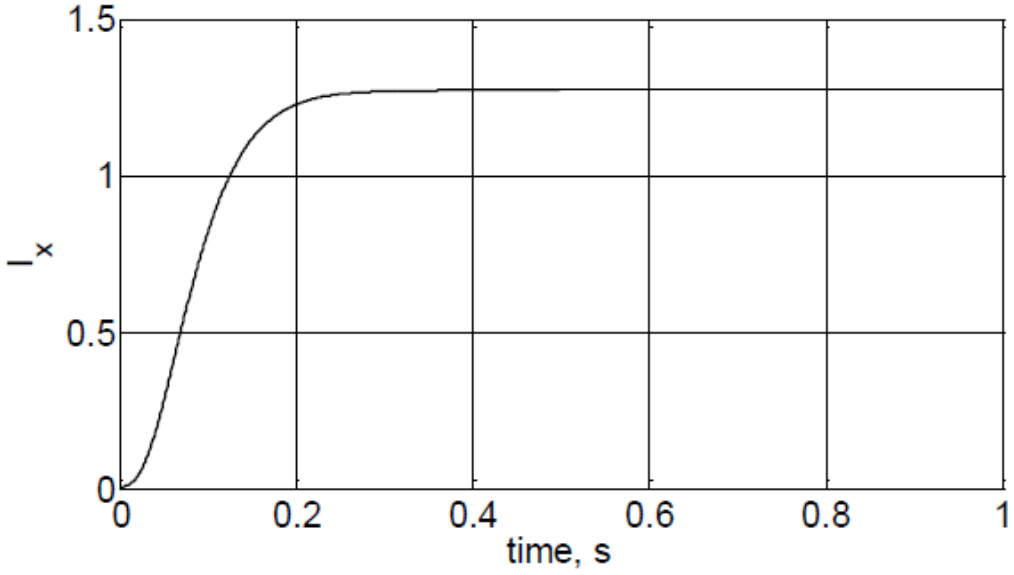
Performance indexes:

$$I_x = \int (x_1^2 + x_2^2)\,\mathrm{dt}$$

In this case the index has as a final value $\boldsymbol{I_x = 1.27}$.

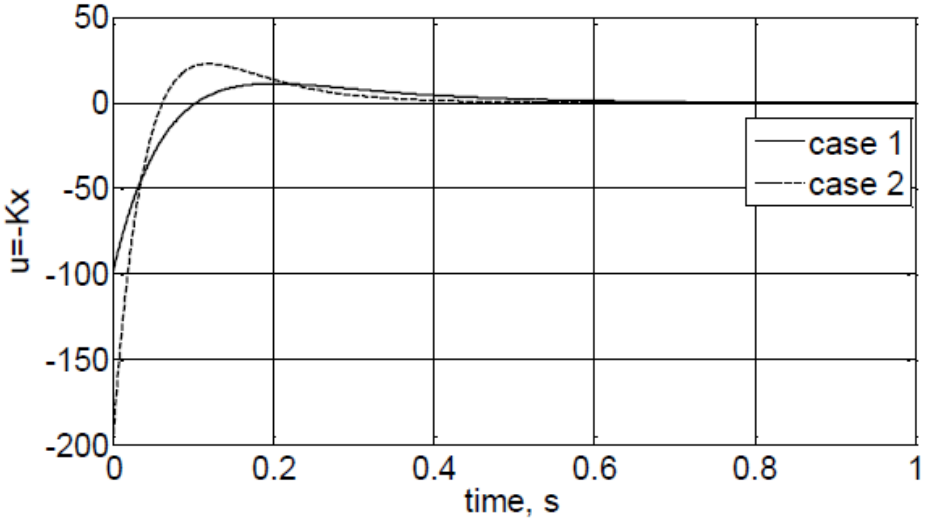






Case 1 vs- Case 2

- **Case 1:** $I_x = 1.40$ (Poles: -10, -11); **Case 2:** $I_x = 1.27$ (Poles: -15,-20)
- Input signal u in **Case 2** is higher Case 1.



- In order to quantify this "energy", we can use a similar index as $I_u = \int (u_1^2 + u_2^2)\,\mathrm{dt}$
- To design a controller that provides the *best* results with respect to some *performance measure* or, alternatively, a controller that minimizes a certain cost function. – Optimal control

Motivation

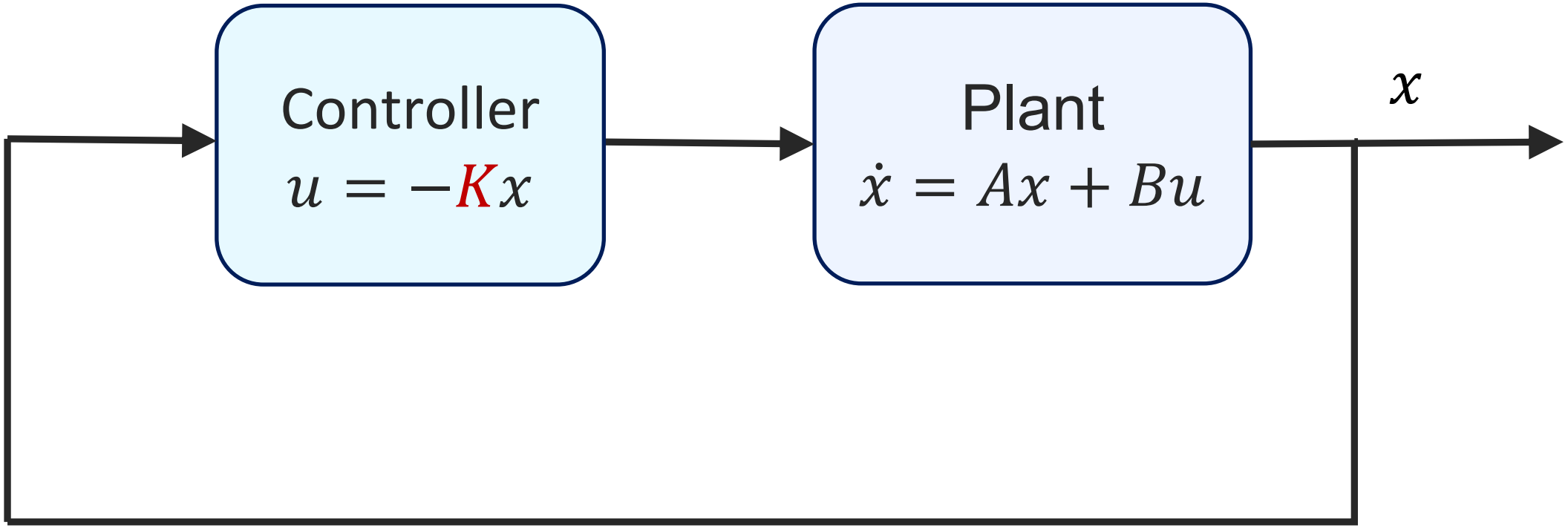


Optimizing the control effort to perform certain tasks is important in certain systems

To design a controller that provides the *best* results with respect to some *performance measure* or, alternatively, a controller that minimizes a certain cost function. – Optimal control

What is Linear Quadratic Regulator?

- Linear Quadratic Regulator (LQR) is an optimization method.
- It is used to **find the gain K** of a **state-feedback control.**



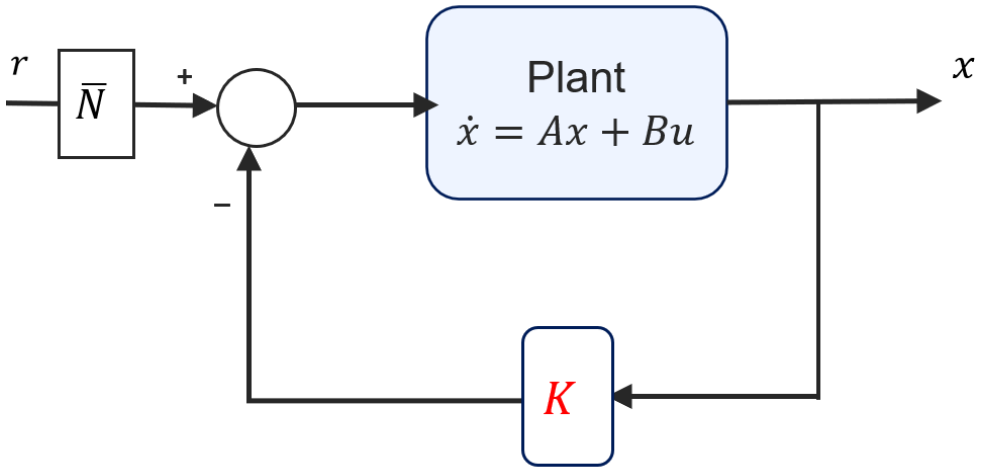
Why LQR?

- **Example**: Control the speed of a vehicle while using very little control (e.g., fuel).
- For this, design a **state feedback controller** that stabilizes the system and drives the state error to zero while using the *least possible control effort*.
- LQR helps us to achieve this goal



Pole placement

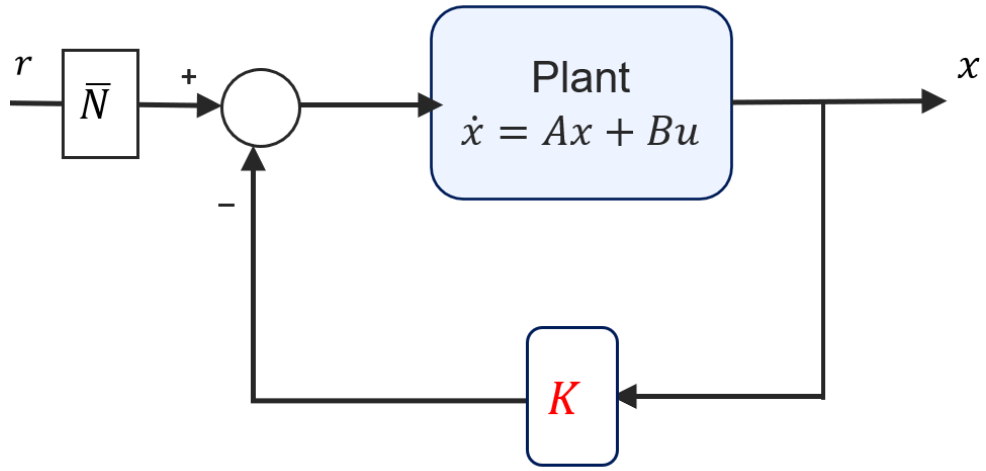
full state feedback



Pole placement: solve for K by choosing pole locations

LQR

full state feedback



LQR: find the optimal K by choosing characteristics, e.g., performance & effort (energy)

LQR Design

- For a system with the state-space model:

$$\dot{x} = Ax + Bu$$

- LQR finds a control law:

$$u = -Kx$$

- That minimizes the cost function

$$J = \int_0^\infty \underbrace{x(t)^T Qx(t)}_{\text{State cost with weighting matrix } Q} + \underbrace{u(t)^T Ru(t)}_{\text{Control cost with weighting matrix } R} dt$$

Performance Effort

- $Q \geq 0, R \geq 0$ are symmetric positive definite matrices.

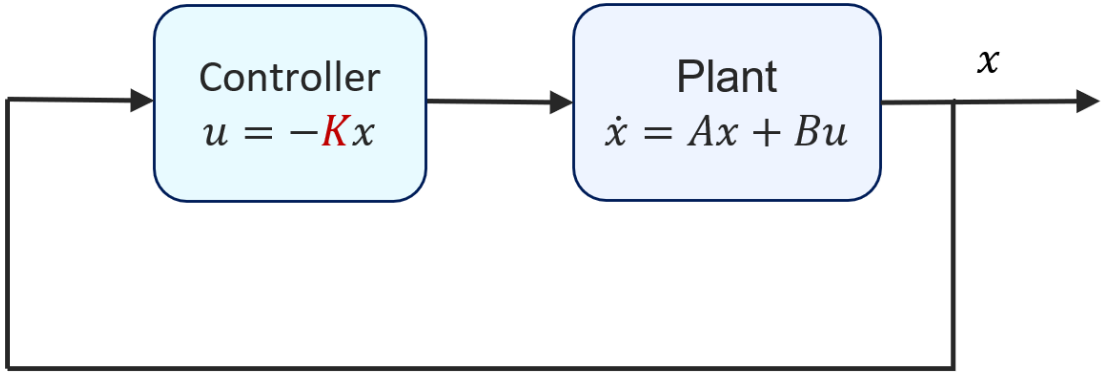
(e.g. diagonal positive)

LQR

- Linear Quadratic Regulator (LQR) is an optimization method.
- It is used to **find the gain K of a state-feedback control.**
- **To Minimize:** $J = \int_0^\infty x(t)^T Qx(t) + u(t)^T Ru(t)dt$

Subject to $\dot{x} = Ax + Bu$

Q and R are symmetric positive definite matrices.



Method 1: Solve LQR problem

- Minimize $J = \int_0^\infty (x^T Q\, x + u^T\, R\, u) dt$, subject to $\dot{x} = Ax + Bu$
- What u minimizes J?
- Introduce $P = P^T$

$$J = x_0^T\, P\, x_0\, - x_0^T\, P\, x_0 + \int_0^\infty (x^T Q\, x + u^T\, R\, u) dt$$

$$= x_0^T\, P\, x_0\, + \int_0^\infty \left[\frac{d}{dt}(x^T P\, x) + x^T Q\, x + u^T\, R\, u\right] dt$$

$\int_0^\infty \left[\frac{d}{dt}(x^T P\, x)\right] dt = [x^T P\, x]_0^\infty = 0 - x_0^T\, P\, x_0, \quad [x^T P\, x]_\infty = 0$ for a stable system.

$$\frac{d}{dt}(x^T P\, x) = \dot{x}^T P\, x + x^T P\, \dot{x}\ = (Ax + Bu)^T P\, x + x^T P (Ax + Bu)$$

Method 1: Solve LQR problem

$$J = x_0^T\, P\, x_0 + \int_0^\infty \left[\frac{d}{dt}(x^T P\, x) + x^T Q\, x + u^T\, Ru\right] dt$$

Insert $\frac{d}{dt}(x^T P\, x) = (Ax + Bu)^T P\, x + x^T P(Ax + Bu)$ to J

$$J = x_0^T\, P\, x_0 + \int_0^\infty [(Ax + Bu)^T P\, x + x^T P(Ax + Bu) + x^T Q\, x + u^T\, R\, u]dt$$

Group $x^T(*)\, x$ terms together

$$J = x_0^T\, P\, x_0 + \int_0^\infty [x^T(A^T P + PA + Q)x + \underline{u^T\, R\, u + x^T PB\, u + u^T B^T P\, x}\,]dt$$

Using $u^T\, R\, u + x^T PB\, u + u^T B^T P\, x = (u + R^{-1}B^T Px)^T R\,(u + R^{-1}B^T Px) - x^T(PBR^{-1}B^T P)x$

$$J = x_0^T\, P\, x_0 + \int_0^\infty [x^T(A^T P + PA + Q - PBR^{-1}B^T P)x + (u + R^{-1}B^T Px)^T R\,(u + R^{-1}B^T Px)]dt$$

Method 1: Solve LQR problem

$$J = x_0^T P x_0 + \int_0^\infty [x^T (A^T P + PA + Q - PBR^{-1}B^T P)x + (u + R^{-1}B^T Px)^T R (u + R^{-1}B^T Px)]dt$$

Let's make this 0 !

$A^T P + PA + Q - PBR^{-1}B^T P = 0$

This is Algebraic Riccati equation

This is 0 if $u = -R^{-1}B^T Px$

$u = -K x$ (full state feedback control)

where $K = R^{-1}B^T P$

- Find P that solves the Algebraic Riccati equation. $A^T P + PA + Q - PBR^{-1}B^T P = 0$
- Find the optimal control gain K. $K = R^{-1}B^T P$

Method 2: Solve LQR - Riccati Equation

Applying the control input $u = -Kx$ to the cost function:

$$J = \int_0^\infty x(t)^T Qx(t) + u(t)^T Ru(t)dt = \int_0^\infty \underbrace{x^T (Q + K^T R K) x}_{\text{Quadratic function}} dt$$

- Define: $P = P^T$, $-\frac{d}{dt}(x^T P x) = -(\dot{x}^T P x + x^T P \dot{x})$ Where P is a real symmetric matrix.
- Make: $-\frac{d}{dt}(x^T P x) = x^T (Q + K^T RK)x$
- Under state-feedback, the state-space model $\dot{x} = (A - BK)x$
- Applying this to equation

$$\begin{aligned} & \dot{x}^T P x + x^T P \dot{x} \\ &= x^T (A - BK)^T P x + x^T P (A - BK) x \quad \text{(group } x^T(*)\, x \text{ terms together)} \\ &= x^T [(A - BK)^T P + P (A - BK)] x \\ &= -x^T (Q + K^T R K) x \end{aligned}$$

Method 2: Finding K using Algebraic Riccati Equation

This is **Riccati equation**:

$$(A - BK)^T P + P(A - BK) = -(Q + K^T R K)$$

It can be shown that if $(A - BK)$ is stable, then we can obtain the **reduced-matrix Riccati equation**:

$$A^T P + PA - PBR^{-1}B^T P + Q = 0$$

The resulting control gain based on this optimization:

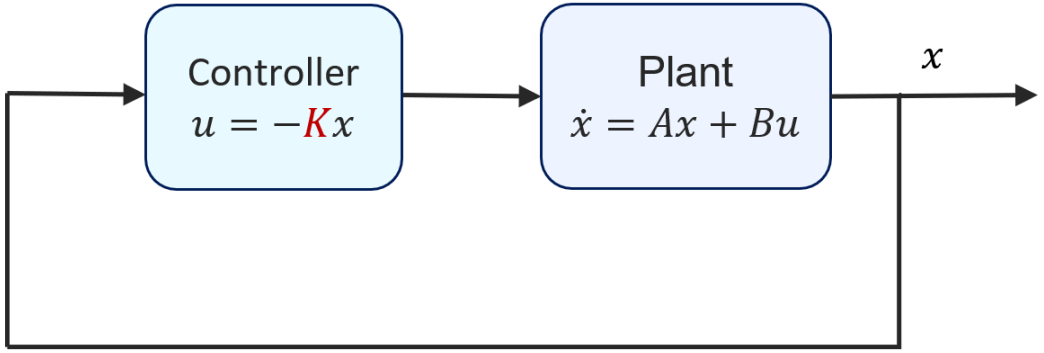
$$K = R^{-1}B^T P$$

- In general, $Q \geq 0, R \geq 0$ are any positive definite matrices. But we will use symmetric diagonal positive definite matrices.

LQR

- Linear Quadratic Regulator (LQR) is an optimization method.
- For a state space system $\dot{x} = Ax + Bu$, it is used to **find the gain**

  $\boldsymbol{K}$ of a **state feedback control** $\boldsymbol{u = -Kx}$.



- **Minimize** $J = \int_0^\infty x(t)^T Qx(t) + u(t)^T Ru(t)dt$

  where Q and R are positive definite matrices.

- Find P that solves the Algebraic Riccati Equation.

$$A^T P + PA + Q - PBR^{-1}B^T P = 0$$

- Find the optimal control gain K.

$$K = R^{-1}B^T P$$

How do I generate the gain K?

In general, it is difficult to solve the Reduced Riccati Equation for high order system. Various software platforms, e.g. Matlab, have tools to generate the control gain..

Matlab: Use the "lqr" command from the Control Systems Toolbox

```
>> Q = eye(2,2);
>> R = 1;
>> K = lqr(A,B,Q,R);
```

How Q and R affect?

$$J = \int_0^{\infty} x(t)^T Q x(t) + u(t)^T R u(t) dt$$

Penalize bad performance

Penalize actuator effort

Low error state is important?

Penalize it by increasing Q

Expensive actuator?

Penalize it by increasing R

How do Q and R weight change K?

Consider the following system:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u$$

$$J = \int_0^\infty x(t)^T Qx(t) + u(t)^T Ru(t)dt$$

To gain an intuition above how Q and R affect the control gain that will be generated

$$Q = \begin{bmatrix}q_1 & 0\\ 0 & q_2\end{bmatrix}$$

$$J = \int_0^\infty \begin{bmatrix}x_1 & x_2\end{bmatrix}\begin{bmatrix}q_1 & 0\\ 0 & q_2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + ru^2\,dt$$

$$K = R^{-1}B^T P$$

$$A^T P + PA - PBR^{-1}B^T P + Q = 0$$

$$J = \int_0^\infty q_1 x_1^2 + q_2 x_2^2 + ru^2\,dt$$

Increasing Q generates a **larger control gain** because LQR needs to "work harder" to reduce the cost function

Increasing R generates a **lower control gain** because LQR needs to "work less" to reduce the cost function.

LQR Control Method – Simple Example

Objective: To gain insight into the underlying theory of optimal control, using a simple example, we will demonstrate how LQR generates control gain K.

Consider the following system:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u$$

With the weighting matrices:

$$Q = \begin{bmatrix}q_1 & 0\\ 0 & q_2\end{bmatrix} \quad R = 1$$

Applying this system to the reduced-order Riccati equation:

$$A^T P + PA - PBR^{-1}B^T P + Q = 0$$

For a system with two states and one input we have:

$$P = \begin{bmatrix}p_{11} & p_{12}\\ p_{12} & p_{22}\end{bmatrix}$$ Real symmetric matrix

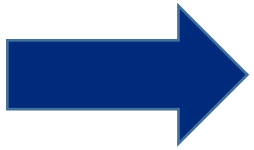
Going through the design

$$A^T P + PA - PBR^{-1}B^T P + Q = 0$$

$$\begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix}\begin{bmatrix}p_{11} & p_{12}\\ p_{12} & p_{22}\end{bmatrix} + \begin{bmatrix}p_{11} & p_{12}\\ p_{12} & p_{22}\end{bmatrix}\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix} - \begin{bmatrix}p_{11} & p_{12}\\ p_{12} & p_{22}\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix} r \begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}p_{11} & p_{12}\\ p_{12} & p_{22}\end{bmatrix} + \begin{bmatrix}q_1 & 0\\ 0 & q_2\end{bmatrix} = 0$$

$$\begin{bmatrix}0 & 0\\ p_{11} & p_{12}\end{bmatrix} + \begin{bmatrix}0 & p_{11}\\ 0 & p_{12}\end{bmatrix} - r\begin{bmatrix}p_{12}\\ p_{22}\end{bmatrix}\begin{bmatrix}p_{12} & p_{22}\end{bmatrix} + \begin{bmatrix}q_1 & 0\\ 0 & q_2\end{bmatrix} = 0$$

$$\begin{bmatrix}0 & p_{11}\\ p_{11} & 2p_{12}\end{bmatrix} - r\begin{bmatrix}p_{12}^2 & p_{12}p_{22}\\ p_{12}p_{22} & p_{22}^2\end{bmatrix} + \begin{bmatrix}q_1 & 0\\ 0 & q_2\end{bmatrix} = 0$$



$$\begin{bmatrix}-p_{12}^2 + q_1 & p_{11} - p_{12}\,p_{22}\\ p_{11} - p_{12}p_{22} & 2p_{12} - p_{22}^2 + q_2\end{bmatrix} = 0$$

Solve for these equations to get matrix P

Find *P* Matrix and Resulting Gain *K*

Solving for the following equations:

$$q_1 - p_{12}^2 = 0$$

$$p_{11} - p_{12}p_{22} = 0$$

$$2p_{12} - p_{22}^2 + q_2 = 0$$

We get

$$P = \begin{bmatrix} \sqrt{2q_1 + \sqrt{q_1}q_2} & \sqrt{q_1} \\ \sqrt{q_1} & \sqrt{2\sqrt{q_1} + q_2} \end{bmatrix}$$

$$K = R^{-1}B^T P$$

$$K = [0 \quad 1]\begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} = [p_{12} \quad p_{22}]$$

$$K = \left[\sqrt{q_1} \quad \sqrt{2\sqrt{q_1} + q_2}\right]$$

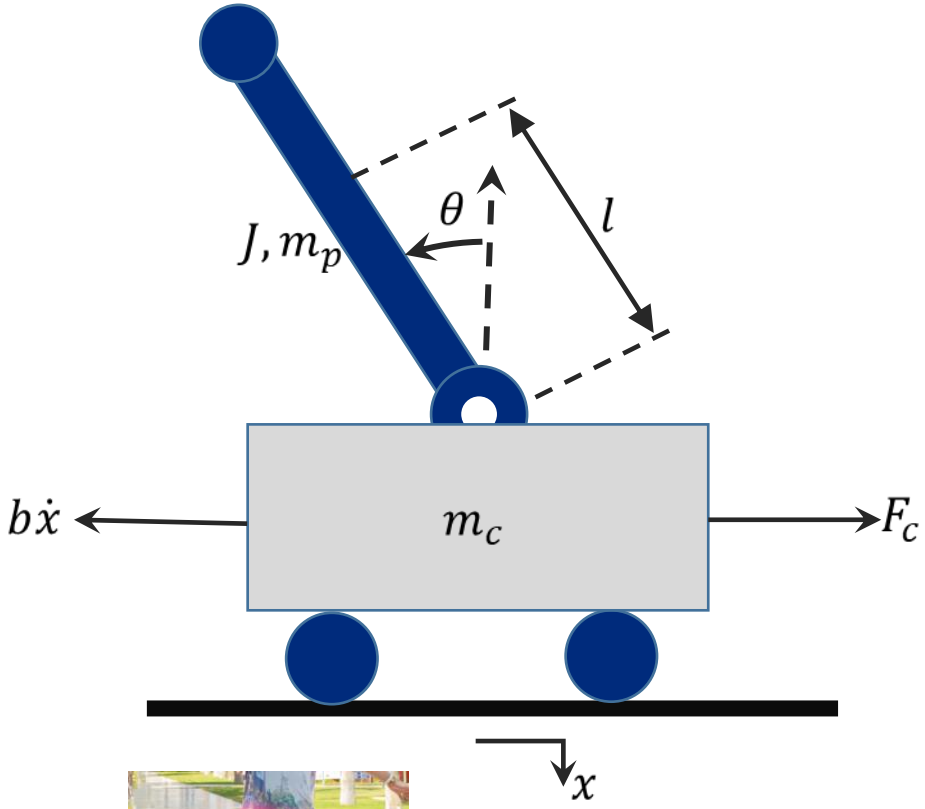
If we choose the following Q matrix, we get the control gain :

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow K = [1 \quad 1.73]$$

Increasing $q_1 = 2$, the control gain increases:

$$Q = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow K = [1.41 \quad 1.96]$$

Example: Linear Inverted Pendulum




Segway

Linear equations of motion:

$$(m_c + m_p)\ddot{x} = F_c - b\,\dot{x} + m_p l\,\ddot{\theta}$$

$$(J + m_p l^2)\ddot{\theta} = m_p l\ddot{x} + m_p g l\theta$$

Define the state and control input:

$$x^T = [x \quad \dot{x} \quad \theta \quad \dot{\theta}] \qquad u = F_c$$

Linear state-space representation (after solving EOMs):

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -(J + m_p l^2)b/J_t & m_p^2 g l^2/J_t & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -m_p l b/J_t & m_p g l(m_c + m_p)/J_t & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ (J + m_p l^2)/J_t \\ 0 \\ (m_p l)/J_t \end{bmatrix} u$$

Where

$$J_t = (J + m_p l^2)(m_c + m_p) - m_p^2 l^2$$

Is this linear inverted pendulum controllable?

Given certain physical parameters:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -6.85 & 2.70 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -16.1 & 29.3 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 50.7 \\ 0 \\ 27.8 \end{bmatrix} u$$

Controllability matrix:

$$C_{ab} = [B \quad AB \quad A^2B \quad A^3B]$$

Rank of controllability matrix:

$$rank(C_{ab}) = 4$$

System is controllable.

LQR Design: Cost Function Analysis

Given the cost function:

$$J = \int_0^\infty x(t)^T Q x(t) + u(t)^T R u(t) dt$$

$$R = r, Q = \begin{bmatrix} q_1 & 0 & 0 & 0 \\ 0 & q_2 & 0 & 0 \\ 0 & 0 & q_3 & 0 \\ 0 & 0 & 0 & q_4 \end{bmatrix}$$

We can evaluate how the Q and R elements affect the control gain:

State-feedback control:

$$u = -Kx = -[k_1 \quad k_2 \quad k_3 \quad k_4]x$$

$$J = \int_0^\infty [x_1 \quad x_2 \quad x_3 \quad x_4] \begin{bmatrix} q_1 & 0 & 0 & 0 \\ 0 & q_2 & 0 & 0 \\ 0 & 0 & q_3 & 0 \\ 0 & 0 & 0 & q_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + ru^2 \, dt$$

$$\Rightarrow \quad J = \int_0^\infty q_1 x_1^2 + q_2 x_2^2 + q_3 x_3^2 + q_4 x_4^2 + ru^2 \, dt$$

State and control input:

$$x^T = [x \quad \dot{x} \quad \theta \quad \dot{\theta}] \qquad u = F_c$$

Designing Gain

$x^T = [x \quad \dot{x} \quad \theta \quad \dot{\theta}]$

$u = F_c$

$u = -[k_1 \quad k_2 \quad k_3 \quad k_4]x$

$$J = \int_0^\infty x(t)^T Qx(t) + u(t)^T Ru(t)dt$$

$$J = \int_0^\infty q_1 x_1^2 + q_2 x_2^2 + q_3 x_3^2 + q_4 x_4^2 + ru^2\, dt$$

$K = R^{-1}B^T P$

Design 1: Choose identity matrix for both Q and R

$$Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R = 1 \quad \Rightarrow \quad K = [-1.00 \quad 2.42 \quad -9.34 \quad -2.03]$$

Design 2: Increasing elements in Q will generate a larger gain values

$$Q = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R = 1 \quad \Rightarrow \quad K = [-1.41 \quad 2.48 \quad -9.91 \quad -2.13]$$

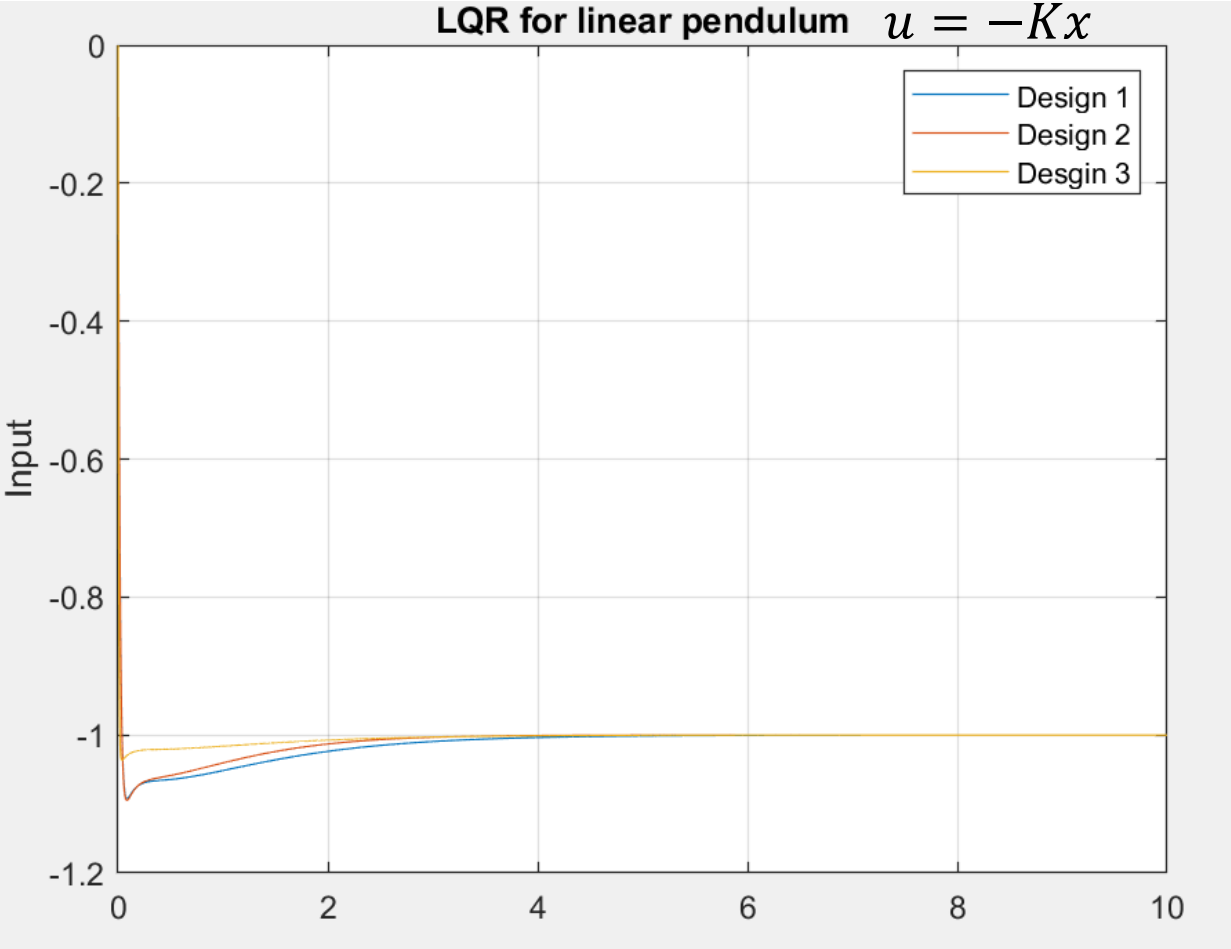
Increasing $q_1 = 2$ generates a **larger control gains** k_1 and k_3 acting on those states

Design 3: Decreasing R forces optimization to find a larger overall control gain.

$$Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R = 0.1 \quad \Rightarrow \quad K = [-3.16 \quad 7.12 \quad -27.69 \quad -6.11]$$

Decreasing R generates an overall **larger control gain**

Simulation with LQR for linear inverted pendulum

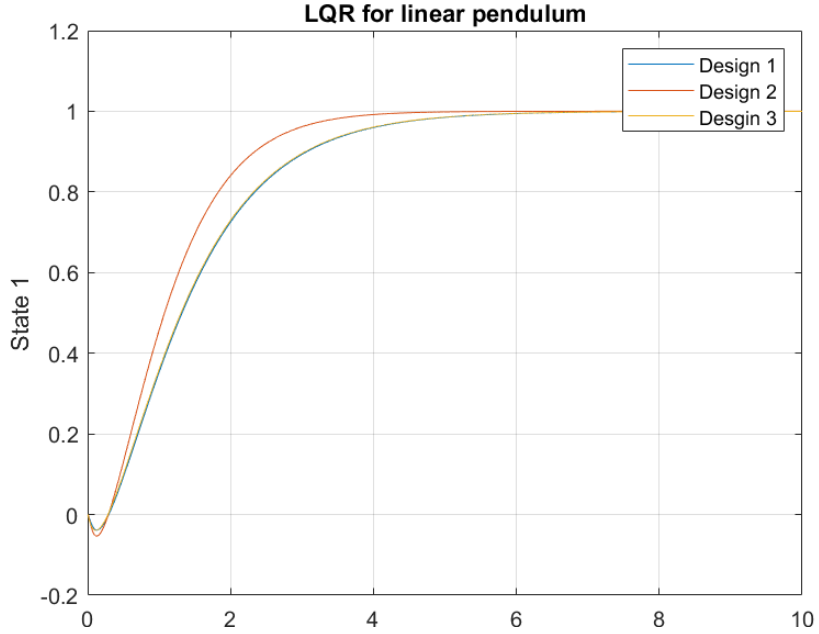


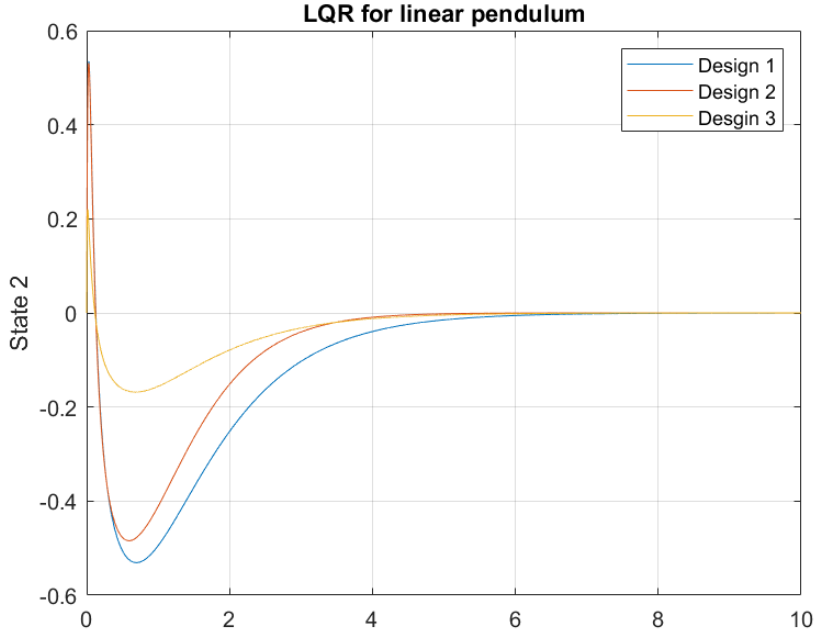
Matlab: L3g_LQR_linear_penduleum.m

Design 1 $Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $R = 1$

Design 2 $Q = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $R = 1$

Design 3 $Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $R = 0.1$

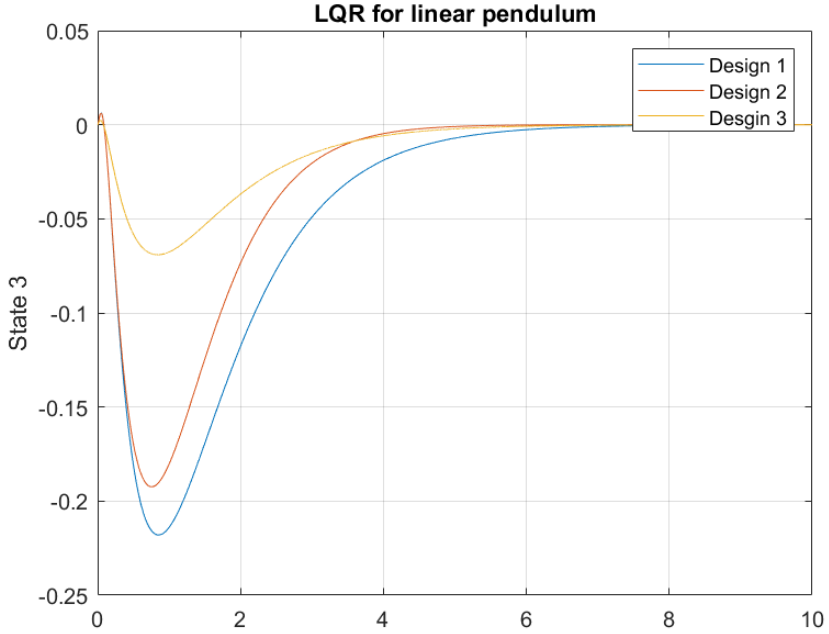


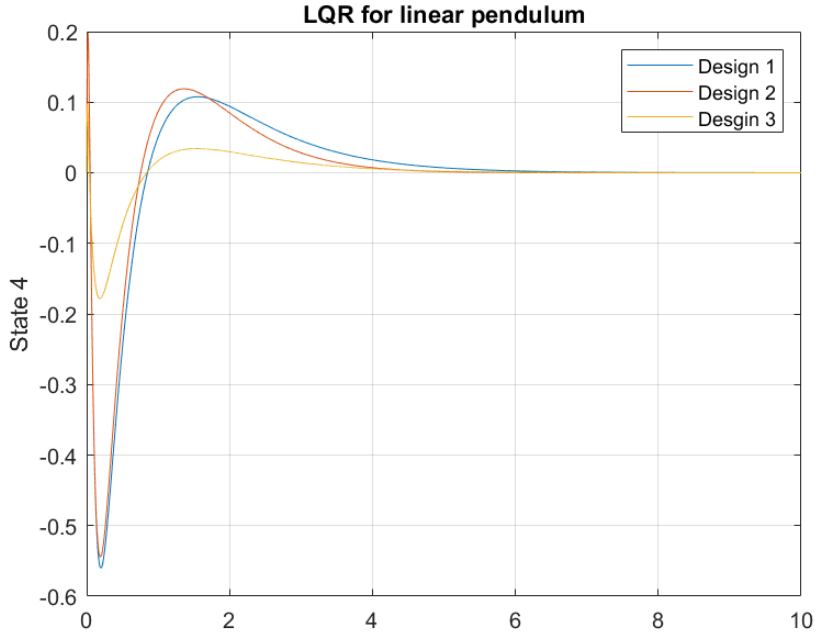


Design 1 $Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $R = 1$

Design 2 $Q = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $R = 1$

Design 3 $Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $R = 0.1$





Bryson's Rule as a first choice

- Choosing Q and R weighting matrices in LQR: Try and Error
- Bryson's rule scales the variables that appear in LQR so that the maximum acceptable value for each term is 1.

Bryson' Rule: A design guideline for LQR.

$$J = \int_0^\infty x(t)^T Qx(t) + u(t)^T Ru(t)dt$$

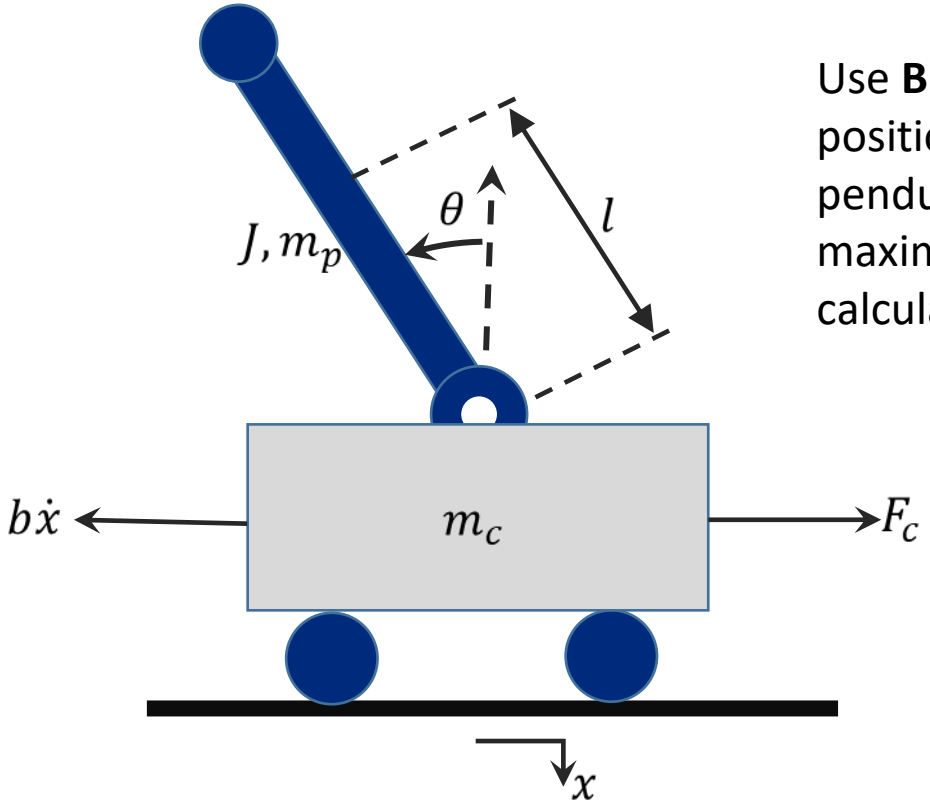
$$Q = \begin{bmatrix} q_{11} & & 0 \\ & \ddots & \\ 0 & & q_{nn} \end{bmatrix} \qquad R = \begin{bmatrix} r_{11} & & 0 \\ & \ddots & \\ 0 & & r_{mm} \end{bmatrix}$$

$$q_{ii} = \frac{1}{\left|x_i^2\right|_{max}} \qquad r_{ii} = \frac{1}{\left|u_j^2\right|_{max}}$$

Where for a system with n states $i = 1 \ldots n$

Where for a system with m inputs $j = 1 \ldots m$

Example: Linear Inverted Pendulum



Use **Bryson's Rule** to find weighting matrices Q and R given the maximum cart position and velocity are 0.1 m and 0.1 m/s, respectively, and the maximum pendulum angle and rate are 10° and 10°/s, respectively. Based on the maximum voltage 10V, the maximum cart force is 4.65 N. Show your calculations and the final Q and R matrices.

Solution: The Q matrix for four state is $x^T = [x \quad \dot{x} \quad \theta \quad \dot{\theta}]$

$$q_{11} = \frac{1}{|x_1^2|_{max}} = \frac{1}{0.1^2} = 100$$

$$q_{22} = \frac{1}{|x_2^2|_{max}} = \frac{1}{0.1^2} = 100$$

$$q_{33} = \frac{1}{|x_3^2|_{max}} = \frac{1}{\left(10\frac{\pi}{180}\right)^2} = 32.8$$

$$q_{44} = q_{33}$$

Example: Linear Inverted Pendulum

Based on the maximum voltage 10V, the maximum cart force is 4.65 N

The R matrix for the input force is $u = F_c$

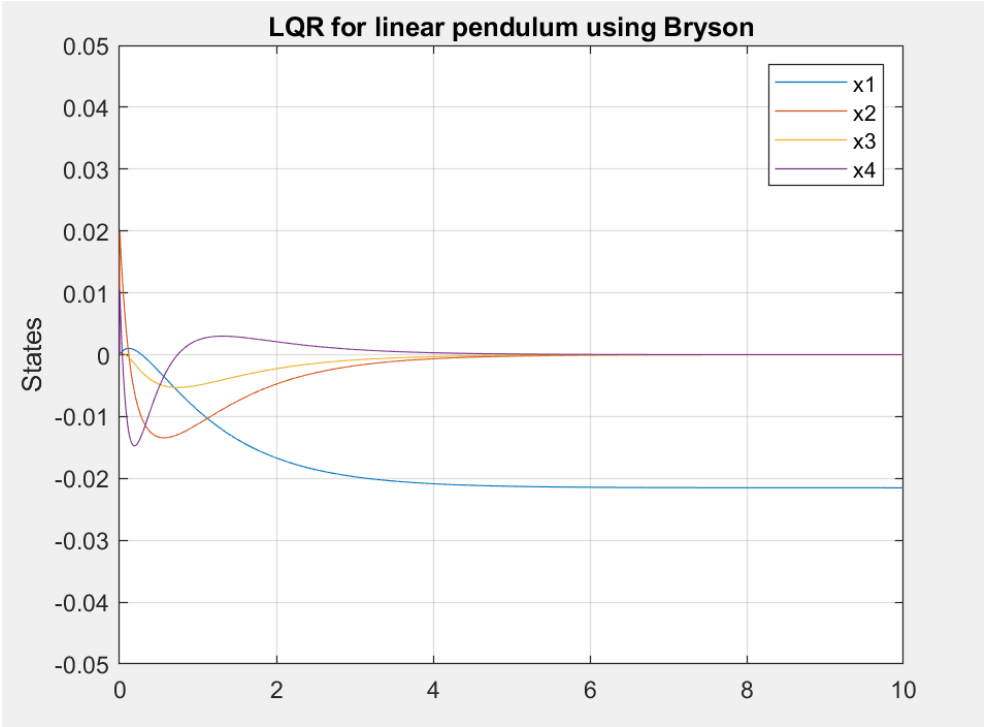
$$R = r_{11} = \frac{1}{|F_c^2|_{max}} = \frac{1}{4.65^2} = 0.0463$$

$$Q = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 100 & 32.8 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 32.8 \end{bmatrix}$$

Using the LQR algorithm in Matlab with these weighting matrices generates the following control gain

$$K = [-46.47 \quad 88.14 \quad -351.85 \quad -71.65\ ]$$

Example: Linear Inverted Pendulum



Conditions for state-feedback control

1. The poles of a system can only be placed arbitrarily via state-feedback if and only if the system is completely state controllable
2. All the states are available, i.e. being measured directly.

Note: If the system is stabilizable, then state-feedback can be used to make the entire system stable

Takeaways

Benefits

- Optimize control effort required
- In software, LQR is a fast way to generate a control gain for closed-loop system

Drawbacks

- May require more control design iterations
  - Not entirely based on time-domain requirements
  - Cannot always predict how Q and R changes gain generated